DiffusionE: Reasoning on Knowledge Graphs via Diffusion-based Graph Neural Networks

Zongsheng Cao
University of Chinese Academy of Sciences
Beijing, China
zongshengcao16@gmail.com

Jing Li
School of Economics and Management, Tsinghua University
Beijing, China
lijing3@sem.tsinghua.edu.cn

Zigan Wang*
Shenzhen International Graduate School, Tsinghua University
Shenzhen, China
School of Economics and Management, Tsinghua University
Beijing, China
wangzigan@sz.tsinghua.edu.cn

Jinliang Li*
School of Economics and Management, Tsinghua University
Beijing, China
jll@tsinghua.edu.cn

ABSTRACT

Graph Neural Networks (GNNs) have demonstrated powerful capabilities in reasoning within Knowledge Graphs (KGs), gathering increasing attention. Our idea stems from the observation that the prior work typically employs hand-designed or sample-designed paradigms in the process of message propagation, engaging a set of adjacent entities at each step of propagation. As a result, such methods struggle with the increasing number of entities involved as propagation steps extend. Moreover, they neglect the message interactions between adjacent entities and propagation relations in KG reasoning, leading to semantic inconsistency during the message aggregation phase. To address these issues, we introduce a novel knowledge graph embedding method through a diffusion process, termed **DiffusionE**. Specifically, we reformulate the message propagation in knowledge reasoning as a diffusion process, regarding the message semantics as the diffusion signal. In this sense, guided by semantic information, messages can be transmitted between nodes effectively and adaptively. Furthermore, the theoretical analysis suggests our method can leverage an adaptive diffusivity for message propagation in the semantic interactions of KGs. It shows that DiffusionE effectively leverages message interactions between entities and propagation relations, ensuring semantic consistency in KG reasoning. Comprehensive experiments reveal that our method attains state-of-the-art performance compared to prior work on several well-established benchmarks.

* Corresponding author.

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CCS CONCEPTS

• **Computing methodologies → Knowledge representation and reasoning**.

KEYWORDS

Knowledge graph, Diffusion process, Graph Neural Networks

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1 INTRODUCTION

Knowledge Graphs (KGs) can represent real-world facts and interrelations between entities in a structured manner [13, 31]. Due to the KG is not complete in practice, KG reasoning aims to infer new facts that are not explicitly presented in the KG by link prediction [15, 36]. This capability has been broadly explored in applications such as predicting drug interactions [16, 19], personalized recommendations [12, 32], and facilitating questions answering [1, 7]. Specifically, a KG reasoning query can be represented as (entity_query, relation_query, ?), then we need to deduce entity_answer from the initial entity_query and relation _query. For instance, given the query (Barack Obama, president of, ?), we can derive the entity_answer is the United States. In this process, KG reasoning entails identifying the desired answer entity for a specific query by leveraging the entities and connections in the KG.

In the past decades, numerous efforts have been made in knowledge graph reasoning methods, which can be categorized into triplet-based models, path-based methods, and Graph Neural Network (GNN)-based approaches. Triplet-based models, such as TransE [6], TorusE [31], and QuatE [40], derive answers by utilizing learned embeddings of entities and relations. Path-based methods, including AnyBURL [3], MultiHop [5], Core [22], and SACN [24], employ logical rules to deduce relational paths from the query entity to

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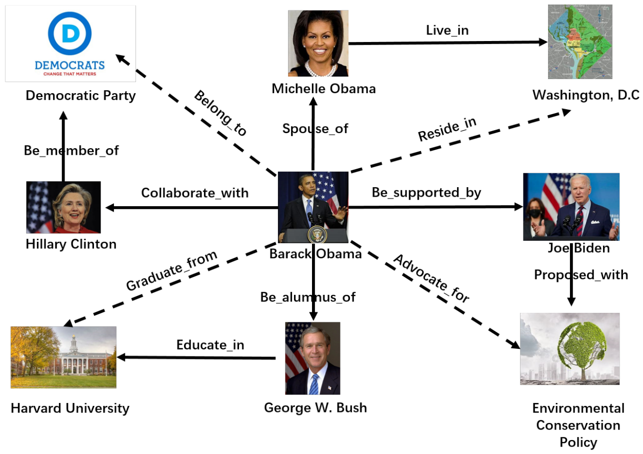


Figure 1: An example of KG reasoning. Given a query (Barack Obama, r,?) the answer depends on the different relation r. Specifically, if r is *Advocate for*, the answer is *Envornmental Conservation Policy* while the answer is *Harvard University* if r is *Graduate from*. In our method, we can aggregate the entities with the guide of semantics gradually, in this way, we can generate the semantic-dependent answer effectively.

the target entity. GNN-based methods enhance entity representations by using GNNs to connect entities with their neighbors [17]. Early models like R-GCN [25] and CompGCN [30] experience oversmoothing issues during message propagation. To this end, recent models such as NBFNet [43] and RED-GNN [41] have addressed this by focusing on the local neighborhoods of the query entity. Despite these advancements, existing methods typically define local neighborhoods based on proximity to the query entity, which may lead to the dilution of distinct semantic features of entities as the number of propagation steps increases.

From the considerations above, it is evident that existing models, particularly GNN-based KG reasoning methods, have achieved significant success but exhibit limitations in fully leveraging the semantics of the query relation when defining relevant local neighborhoods. Specifically, these methods pay less attention to the query relation, hence they often include many extraneous entities and facts, thereby increasing computational overhead without correspondingly enhancing performance. For instance, as illustrated in Fig. 1, the most relevant evidence for the query (Barack Obama, Advocate_for, ?) includes the facts such as (Barack Obama, Be_supported_by, Joe Biden) and (Joe Biden, Proposed_with, Environmental Conservation Policy). Conversely, for the query (Barack Obama, Graduate_from, ?), the optimal response is derived from the subgraph featuring (Barack Obama, Be_alumnus_of, George W. Bush) and (George W. Bush, Educate_in, Harvard University). In light of these observations, two critical challenges arise for KG reasoning: *(1) How can local neighborhoods be dynamically tailored for different query relations? (2) How can message passing effectively capture semantic correlations inherent in the KG?*

To this end, inspired by the diffusion process[1], we propose a novel framework to model the message propagation in KG reasoning, termed **DiffussionE**. Specifically, to tackle challenge (1), we introduce a wave-based propagation [9] module, where the message spreads from the previous wave to the next one, prioritizing crucial nodes first and progressively widening the propagation scope. This method effectively facilitates message transmission without an explosive increase in involved entities. To address the challenge (2), drawing inspiration from the energy-based model in the diffusion process, we introduce a semantics-dependent flowing module, which can capture semantic correlations and distinctions between entities. Moreover, we show the advantage of adaptive diffusivity in the message propagation. On top of this, we theoretically highlight that DiffusionE can leverage interactions between entities and propagation relations to ensure semantic consistency. Extensive experiments on several well-established benchmarks demonstrate the effectiveness and efficiency of our method.

In a nutshell, the contributions of our method are summarized as three-fold:

- We propose a novel KG reasoning method, which reformulates the message propagation as the diffusion process. Specifically, with the assistance of wave-based propagation and semantics-dependent flowing modules, we take the early trial to implement semantic-based information adaptively aggregated messages within a unified framework from a semantic interaction perspective.
- We show the advantage of adaptive diffusivity in knowledge propagation, and our method can realize this goal. In this sense, we theoretically demonstrate that DiffusionE can preserve semantic consistency in message propagation.
- Experiments show that the proposed method achieves state-of-the-art performance in both transductive and inductive KG reasoning scenarios in several datasets.

2 RELATED WORK

2.1 GNN for KG reasoning

The development of Graph Neural Networks (GNNs) has significantly transformed the approach to modeling graph-structured data, particularly in the realm of knowledge graph reasoning. This transformation is marked by widespread interest and utilization, as evidenced by seminal works [10, 17]. Studies such as [28, 30, 41, 43] illustrate the application of the message propagation framework, which facilitates dynamic interactions among entities to enhance KG reasoning [10, 17]. Within this framework, messages at the l-th propagation step are computed as defined by the set $\mathcal{F}^{(l)} = \{(e_s, r, e_o) \in \mathcal{F} | e_s \in \mathcal{E}^{(l-1)}, r \in \mathcal{R}, e_o \in \mathcal{E}^{(l)}\}$. Each message $m^{(l)}(e_s, r, e_o)$ is derived using the function $MESS(\boldsymbol{h}_{e_s}^{l-1}, \boldsymbol{h}_r^{(l)})$, which leverages the respective embeddings of entities and relations, $\boldsymbol{h}_{e_s}^{l-1}$ and $\boldsymbol{h}_r^{(l)}$.

Regarding GNN-based KG reasoning methods, there exist variations in the sets of entities involved in the propagation process, which can be broadly categorized into three classes based on their approach to defining the propagation range: 1) Full propagation

[1]Diffusion here refer to the heat-based diffusion.

methods, such as R-GCN [25] and CompGCN [30], involve all entities in the propagation. These methods face challenges related to high memory demands and the potential for over-smoothing when all entities are included, which typically limits the number of feasible propagation steps. 2) Progressive propagation methods, exemplified by RED-GNN [41] and NBFNet [43], begin with a specific query entity e_q and incrementally expand through its L-hop neighborhood. Initially, $\mathcal{E}^{(0)} = \{e_q\}$, with each subsequent set $\mathcal{E}^{(l)}$ including entities within the expanding neighborhood, $\mathcal{E}^{(l)} = \bigcup_{e \in \mathcal{E}^{(l-1)}} \mathcal{N}(e)$, where $\mathcal{N}(e)$ denotes the 1-hop neighbors. 3) Constrained propagation methods, such as GraIL [28] and CoMPILE [20], employ a restriction on propagation to specific subgraphs. These subgraphs differ for each entity pair, allowing for the aggregation of semantic messages from a subgraph perspective. However, this approach can pose computational challenges in large-scale KGs, as the computational demands increase due to the varying subgraphs.

2.2 Sampling methods for GNN

Sampling methodologies play a crucial role in guiding and refining entity selection during the propagation process in the application of GNNs, particularly for homogeneous graphs. These techniques are essential for enhancing scalability and can be broadly classified into the following three main categories. (a) Node-wise Sampling Methods: Models such as GraphSAGE [11] and PASS [37] selectively sample K entities from the neighboring set, $\mathcal{N}(e)$, for each entity e during each propagation step, focusing on individual nodes. (b) Layer-wise Sampling Methods: Represented by FastGCN [2], Adaptive-GCN [14], and LADIES [44], these methods regulate the number of entities sampled at each layer to K, thus controlling the total number of entities involved in subsequent propagation layers. (c) Subgraph Sampling Methods: Notable methods such as Cluster-GCN [4], GraphSAINT [39], and ShadowGNN [38] focus on extracting localized subgraphs related to a query entity. These methods aim to provide a more focused view of the graph by leveraging the relevance of the sampled subgraphs to the query.

Additionally, several innovative models have been developed to enhance KG reasoning [8, 34]. Among these, RS-GCN [8] introduces a node-wise sampling method specifically tailored to optimize neighbor selection during the propagation phases of R-GCN and CompGCN. This strategy is crucial for reducing memory usage and enhancing the efficiency of large-scale KG operations. AdaProp [42] proposes an adaptive propagation path to facilitate the KG reasoning. Another notable model, DPMPN [34], merges the benefits of a full propagation GNN, adept at capturing global features, with the targeted efficiency of a pruned GNN. This approach utilizes a layer-wise sampling method where pruning is strategically applied, thus significantly improving memory efficiency and scalability in KG reasoning.

Despite the remarkable success, the previous work neglects the semantic-based propagation process, hence leading to the suboptimal performance. In this paper, we propose a new KG reasoning method inspired by the difussion process, which provides a feasible direction to study KG reasoning.

3 PREMILARY

A diffusion process in this paper refers to the concept of heat diffusion on the Riemannian manifold [23]. This process employs an anisotropic diffusion mechanism, which efficiently generates instance representations through the dynamic flow of information. We represent the state of any instance at time t and location i with a vector-valued function $z_i(t) : [0, \infty) \to \mathbb{R}^d$. This anisotropic diffusion is mathematically modeled by a partial differential equation (PDE), equipped with specific boundary conditions, that outlines the temporal evolution of instance representations.

$$\frac{\partial \mathbf{Z}(t)}{\partial t} = \nabla^*(\mathrm{S}(\mathbf{Z}(t), t) \odot \nabla \mathbf{Z}(t)), \quad \text{s. t. } \mathbf{Z}(0) = [\mathbf{x}_i]_{i=1}^N, \quad t \geq 0 \tag{1}$$

In Eq. 1, we define $Z(t) = [z(t)]_{i=1}^N \in \mathbb{R}^{N \times d}$, where $\odot$ denotes the Hadamard product. The function $S(Z(t), t) : \mathbb{R}^{N \times d} \times [0, \infty) \to [0, 1]^{N \times N}$ represents the diffusivity coefficient, which adjusts the diffusion intensity between any two instances at a given time t, based on their current states. The gradient operator ∇ measures the state differences between source and target instances, specifically, $(\nabla Z(t))_{ij} = z_j(t) - z_i(t)$ for distinct locations i and j. Conversely, the divergence operator ∇^* integrates these differences to assess the overall information flow at a location. It is computed as follows:

$$(\nabla^*)_i = \sum_{j=1}^{N} S_{ij}(Z(t), t)(\nabla Z(t))_{ij}.$$

It should be emphasized that both operators are defined within a discrete domain encompassing N locations. The physical interpretation of Eq. 1 suggests that the temporal variation of heat at any given location i corresponds directly to the spatial influx of heat at that location.

$$\frac{\partial \mathbf{z}_i(t)}{\partial t} = \sum_{j=1}^{N} \mathrm{S}_{ij}(\mathbf{Z}(t), t)\left(\mathbf{z}_j(t) - \mathbf{z}_i(t)\right) \tag{2}$$

The diffusivity coefficient, as described in Eq.1, is pivotal in determining the rate of heat dissemination across the space, serving as a crucial metric [23]. In Eq.2, $S(Z(t), t)$ plays a key role in regulating the flow of information between instances and guiding the evolution of their states, offering considerable flexibility in its definition. A straightforward implementation might define $S(Z(t), t)$ as an identity matrix, effectively limiting feature propagation to self-loops only. This approach essentially transforms the model into a multi-layer perceptron (MLP), where each instance is processed independently. Conversely, when a predefined graph structure is available, $S(Z(t), t)$ can be configured to reflect this structure, though this configuration restricts the information flow to the immediate neighbors in the graph. Ideally, $S(Z(t), t)$ would allow non-zero values across any pair (i, j) and dynamically evolve over time, enabling instances at each layer to propagate information adaptively and efficiently to all others, thereby facilitating an extensive exchange of information among instances.

4 THE PROPOSED METHOD

Problem formulation. Let $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{F}, \mathcal{Q})$ represent an instance of a KG. Here, $\mathcal{E}$ and $\mathcal{R}$ denote the sets of entities and relations, respectively. The set of fact triplets is denoted as $\mathcal{F}$ =

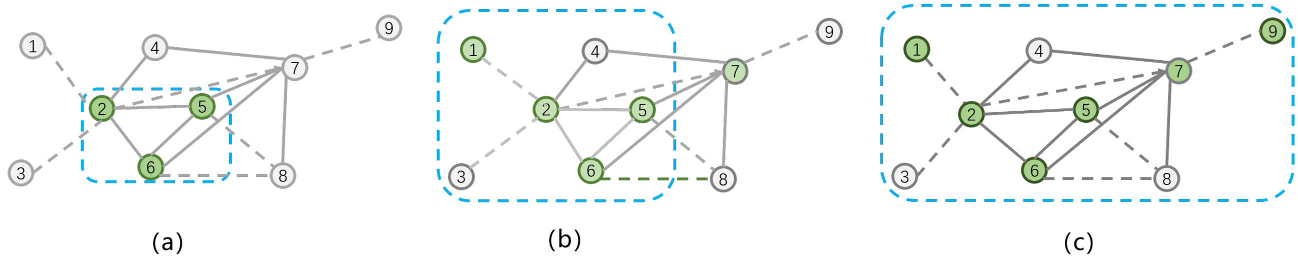


Figure 2: The illustration of the message propagation. From (a) to (b) and (c), the semantics-dependent diffusion module can capture semantic correlations and distinctions between entities. Meanwhile, the message spreads from the previous wave to the next one, prioritizing crucial nodes first and progressively widening the propagation scope.

$\{(e_s, r, e_o) | e_s, e_o \in \mathcal{E}, r \in \mathcal{R}\}$, where each triplet consists of a subject entity e_s, a relation r, and an object entity e_o. Additionally, $Q = \{(e_q, r_q, e_a) | e_q, e_a \in \mathcal{E}, r_q \in \mathcal{R}\}$ represents the set of query triplets, where each triplet contains an incomplete component, indicated by a question mark, and a target answer entity e_a. The goal of the reasoning task is to learn a function that predicts the answer entity $e_a \in \mathcal{E}$ for each query $(e_q, r_q, ?)$ based on the available fact triplets $\mathcal{F}$. In other words, the objective is to develop a model that can infer missing information in the query based on the existing knowledge graph facts.

Diffusion Path. In this part, the primary objective is to design an enhanced propagation path, $\widetilde{\mathcal{G}}^L$, that enables the model, optimized with appropriate parameters, to accurately predict the target answer entity e_a for each query $(e_q, r_q, ?)$. Notably, previous methods have often overlooked the semantic correlations and distinctions among entities during the message propagation process. To address this shortcoming, we leverage a diffusion process to dynamically adapt the propagation paths, as illustrated in Figure 2. This strategy facilitates the gradual dissemination and transfer of information while accounting for the semantic relationships among entities.

$$\widehat{\mathcal{G}}^L_{e_q} = \left\{\mathcal{E}^0_{e_q}, \ldots, \mathcal{E}^\ell_{e_q}, \ldots, \mathcal{E}^L_{e_q}\right\},$$
$$\text{s.t.} \quad \mathcal{E}^\ell_{e_q} = \begin{cases} \{e_q\} & \ell = 0 \\ diff\left(\mathcal{E}^{\ell-1}_{e_q}\right) & \ell = 1 \ldots L \end{cases} \tag{3}$$

The diffusion process commences with a specific query entity, denoted as e_q. Entities residing within $\mathcal{E}^\ell_{e_q}$ are meticulously selected through an adaptive sampling mechanism, encompassing a subset of $\mathcal{E}^{\ell-1}_{e_q}$ and its immediate neighbors, under the influence of the diffusion process. However, this approach is accompanied by a duo of inherent challenges: (1) The diffusion process inherently exhibits a gradual and incremental expansion, proliferating across an increasingly vast spatial domain. (2) A notable variance is observed in the diffusion rates among neighboring nodes, a phenomenon intricately linked with the semantic relationships interconnecting the nodes.

To this end, we introduce a new propagation paradigm with the assistance of the diffusion process, which can adaptively select relevant entities into the propagation path based on the given query. To solve the first challenge, we design a wave-based sampling module [9], which can reduce the number of involved entities and preserve the layer-wise connections and we will gradually expand the wave area to get more and more messages. Notice that the efficiency of diffusion during propagation is greatly influenced by the transmission pathway, such as the disparity in thermal conductivity between the two components. For the second challenge, Inspired by the aforementioned phenomenon, we propose a semantics-dependent sampling distribution, which is jointly optimized with the model parameters, to select entities that are semantically relevant to the query relation.

Motivated by the analogous phenomenon of wave propagation in water, where subsequent waves are generated iteratively from their predecessors, it is mathematically represented by the constraint

$$\mathcal{E}^0_{e_q} \subseteq \mathcal{E}^1_{e_q} \subseteq \cdots \subseteq \mathcal{E}^L_{e_q}$$

ensuring a coherent retention of entities from previous steps, while also facilitating incremental sampling from newly encountered entities. This methodology aims to preclude the omission of potential target entities that exhibit promising attributes.

Wave-based Propagation. Consider the set of entities at the $(\ell-1)$-th step, denoted as $\mathcal{E}^{\ell-1}_{e_q}$. Let the entities designated for sampling in the ℓ-th step be symbolized as $\overline{\mathcal{E}}^{\ell-1}_{e_q}$. Essentially, all adjacent entities expressed as $\mathcal{N}(\mathcal{E}^{\ell-1}_{e_q}) = \cap_{e \in \mathcal{E}^{\ell-1}_{e_q}} \mathcal{N}(e)$, are conceived as viable candidates for the sampling procedure. However, to ensure the sustained preservation of $\mathcal{E}^{\ell-1}_{e_q}$ in succeeding sampling steps, the focus is strategically shifted towards the newly encountered entities, hence identifying them as the prime candidates for sampling, denoted as, *i.e.*,

$$\overline{\mathcal{E}}^\ell_{e_q} := \text{DIFF}\left(\mathcal{E}^{\ell-1}_{e_q}\right) = \mathcal{N}\left(\mathcal{E}^{\ell-1}_{e_q}\right) \setminus \cup^{\ell-1}_{k=n_s} \mathcal{E}^k_{e_q}. \tag{4}$$

where n_s $(1 \leq n_s \leq \ell - 1)$ denotes a specific layer, which controls the propagation range. In this way, influenced by wave-based propagation, it allows for the nuanced incorporation of newly discovered entities while maintaining a robust continuity with previously identified entities, ensuring a comprehensive and dynamic sampling process.

Observe that within the diffusion process, heat energy disseminates unevenly among points, flowing more profusely where temperature differences are more pronounced. A parallel can be drawn with semantic flow processes, where adjacent entities marked by more substantial semantic discrepancies tend to impart more profound informational insights compared to those characterized by minimal differences. Inspired by this, an energy module is introduced to quantify the semantic significance of entities meticulously.

This allows for the sampling of $K \ll |\overline{\mathcal{E}}^t_{e_q}|$ entities, executed without replacement from the candidate set $\overline{\mathcal{E}}^l_{e_q}$, subsequent to a systematic ranking based on their semantic importance. This innovative approach, inspired by natural thermal diffusion processes, aims to optimize the selection of entities by prioritizing those that are semantically more impactful and conducive to the enhancement of informational richness within the diffusion process.

Semantics-dependent Diffusion. In the context of GNN-based KG reasoning methodologies, the entity representations at the termination of the propagation sequence, denoted as $\boldsymbol{h}^L_{e_o}$, serve a pivotal role in ascertaining the plausibility of entity e_o being identified as the target answer entity. This process is inspired by the integration of an energy function within the diffusion process, prompting the proposal of a semantic module. This module is ingeniously crafted to learn and elucidate the entity representations, subsequently indicating their relevance concerning the given query $(e_q, r_q, ?)$.

Specifically, to meticulously gauge the semantic relevance and inherent variability, inspired by the diffusion process, we utilize a linear mapping function to achieve this, expressed as $g(\boldsymbol{h}^l_e; \theta^l)$, replete with parameters $\boldsymbol{\theta}^l \in \mathbb{R}^d$, where l is the layer. In this way, it can facilitate the process of sampling, executed according to a meticulously constructed probability distribution, embodying the nuanced semantic intricacies and relevance of the entities in relation to the propagated query. To this end, we sample entities in the message propagation according to the probability distribution:

$$p^\ell(e) := \frac{\exp\left(f\left(g\left(\boldsymbol{h}^\ell_e; \boldsymbol{\theta}^\ell\right) - g\left(\boldsymbol{h}^\ell_{e_q}; \boldsymbol{\theta}^\ell\right)\right)/\tau\right)}{\sum_{e' \in \overline{\mathcal{E}}^\ell_{e'}} \exp\left(f\left(g\left(\boldsymbol{h}^\ell_{e'}; \boldsymbol{\theta}^\ell\right) - g\left(\boldsymbol{h}^\ell_{e_q}; \boldsymbol{\theta}^\ell\right)\right)/\tau\right)} \tag{5}$$

where the temperature value $\tau > 0$ and we select the top-K candidate entities by calculate $p^\ell(e)$ in $\overline{\mathcal{E}}^\ell_{e_q}$ and denote the set of them as $\widetilde{\mathcal{E}}^\ell_{e_q}$. f is a non-negative and decreasing function such as ReLU function on certain intervals for $g(\cdot;\cdot)$. For more mathematical properties and explanations refer to Theorem.1.

The sampled entities at ℓ-th step together with entities in the $(\ell-1)$-th step constitute the involved entities at the l step, *i.e.*,

$$\mathcal{E}^\ell_{e_q} = \mathcal{E}^{\ell-1}_{e_q} \cup \left(\widetilde{\mathcal{E}}^\ell_{e_q}\right) \tag{6}$$

Message Aggregation. In the process of message aggregation, the foundational step involves the obtaining of the neighborhood entity set, accomplished through a sophisticated diffusion process. Subsequent to acquiring the neighborhood entities, the algorithm proceeds to the crucial phase of message representation calculation. In this way, consider a neighborhood entity set, denoted by $\mathcal{N}$, which has been ascertained through the diffusion mechanism. Given the query $(e_q, r_q, ?)$ and edges $\mathcal{F} = (e_s, r, e_o)$ in the diffusion process, the objective then metamorphoses into the calculation of an aggregated message representation, delineated as follows:

$$\boldsymbol{m}^\ell_{e_s,r,e_o} = \mathrm{MESS}(\boldsymbol{h}^{\ell-1}_{e_s}, \boldsymbol{h}^{\ell-1}_{e_o}, \boldsymbol{h}^\ell_r, \boldsymbol{h}^\ell_{r_q}) \tag{7}$$

where MESS$(\cdot)$ can be modelled by add function and entities $e_o \in \mathcal{N}(\mathcal{E}^\ell_{e_q})$. On top of this, we conduct the message aggregation process as follows:

$$\boldsymbol{h}^\ell_{\boldsymbol{e}_o} = s\left(\mathrm{AGG}\left(\boldsymbol{m}^\ell_{(\boldsymbol{e}_s,r,\boldsymbol{e}_o)}, (e_s, r, \boldsymbol{e}_o) \in \mathcal{F}^\ell\right)\right) \tag{8}$$

Algorithm 1: DiffusionE: learning adaptive propagation path.

Input: query $(e_q, r_q, ?), \mathcal{E}_{e_q} = \{e_q\}$, steps L, number of sampled entities K, and functions $MESS(\cdot)$ and $AGG(\cdot)$.

Output: The probability being the target answer of $e_o \in \mathcal{E}_{e_q,r_q}$

for $l = 1, \cdots, L$ **do**

- Get the adjacent entities $\mathcal{N}(\mathcal{E}^{l-1}_{e_q}) = \cap_{e \in \mathcal{E}^{l-1}_{e_q}} \mathcal{N}(e)$, the newly-diffusion entities $\overline{\mathcal{E}}^l_{e_q} = \mathcal{N}(\mathcal{E}^{l-1}_{e_q}) \backslash \mathcal{E}^{l-1}_{e_q}$, and edge $\mathcal{F}^l = \{(e_s, r, e_o) | e_s \in \mathcal{E}^{l-1}_{e_q}, e_o \in \mathcal{N}(\mathcal{E}^{l-1}_{e_q})\}$
- Obtain $\boldsymbol{m}^l_{e_s,r,r_o} = MESS(\boldsymbol{h}^{l-1}_{e_s}, \boldsymbol{h}^{l-1}_{e_o}, \boldsymbol{h}^l_r, \boldsymbol{h}^l_{r_q})$ for entities $e_o \in \mathcal{N}(\mathcal{E}^{l-1}_{e_q})$
- Obtain $\boldsymbol{h}^\ell_{\boldsymbol{e}^\ell_o} := \delta\left(\mathrm{AGG}\left(\boldsymbol{m}_{(\boldsymbol{e}_s,r,\boldsymbol{e}_o)}, (e_s, r, \boldsymbol{e}_o) \in \mathcal{F}^\ell\right)\right)$ for entities $e_o \in \mathcal{N}(\mathcal{E}^{l-1}_{e_q})$
- Generate the $\widetilde{\mathcal{E}}^\ell_{e_q}$ by Eq.(5)
- Update propagation path: $\mathcal{E}^\ell_{e_q} = \mathcal{E}^{\ell-1}_{e_q,r_q} \cup \widetilde{\mathcal{E}}^\ell_{e_q}$

Return ϕ_{e_o} for each $e_o \in \mathcal{E}^L_{e_q}$.

where AGG$(\cdot)$ can be modeled by function Mean$(\cdot)$, and s is an activation function such as the sigmoid function.

Loss Function. By evaluating the diffusion rate among different entities in the above, our method can effectively optimize the message propagation paths, better capture the semantic correlation among entities, as well as reduce the redundant paths, and increase the accuracy of inference. In this way, we then denote $\mathcal{L}$ as the loss function instantiated on each instance, employing binary cross-entropy loss across all entities $e_o \in \mathcal{E}^L_{e_q}$, delineated as:

$$\mathcal{L} = -\sum_{e_o \in \mathcal{E}^L_{e_q}} y_{e_o} \log\left(\phi_{\boldsymbol{e}_o}\right) + \left(1 - y_{e_o}\right) \log\left(1 - \phi_{e_o}\right) \tag{9}$$

where ϕ_{e_o} denotes the likelihood score for the plausibility of e_o being identified as the target answer entity and it can be modeled by linear transformation. e_a is the answer entity . Concurrently, the label y_{e_o} is ascribed a value of 1 if the condition $e_o = e_a$ holds true, otherwise, it is allocated a value of 0. Moreover, the overall algorithm can refer to Algorithm 1.

Remark 1. *Algorithm 1 presents a significant advancement by utilizing a dynamic propagation path instead of a predefined static path. This adaptability in propagation is guided by the wave-based propagation mechanism and a diffusion-based strategy, allowing the path to be recalibrated and updated at each step based on the evolving representations of entities.*

5 THEORETICAL ANALYSIS

In the above, we delineate the rationale behind DiffusionE in sampling neighborhood entities, and then we show the advantage of diffusivity in our method as follows.

Intrinsic adaptive diffusivity of DiffusionE. Recall the diffusion process mentioned in Section 3, to solve Eq.(2), such a diffusion process can guide the model to aggregate the semantic-relevant message from other instances at every layer for learning informative instance representations. To this end, we can use numerical methods to solve the continuous dynamics in Eq.(2), e.g., the explicit Euler scheme involving finite differences with step size τ as follows:

$$z_i^{(k+1)} = \left(1 - \tau \sum_{j=1}^{N} S_{ij}^{(k)}\right) z_i^{(k)} + \tau \sum_{j=1}^{N} S_{ij}^{(k)} z_j^{(k)}. \tag{10}$$

The numerical iteration can stably converge for $\tau \in (0, 1)$. We can adopt the state after a finite number K of propagation steps and use it for final predictions, i.e., $\hat{y}_i = MLP(z_i^{(K)})$.

On the basis of this, we then introduce an energy function, designed to quantitatively assess the anticipated quality of instance states at a specific iteration denoted by k as follows:

$$E(Z, k; \delta) = \|Z - Z^{(k)}\|_F^2 + \lambda \sum_{i,j} \delta \|z_i - z_j\|_2^2 \tag{11}$$

where $\delta : \mathbb{R}^+ \rightarrow \mathbb{R}$ is a non-decreasing and concave function on a certain interval. In this way, we have the following theorem:

Theorem 1. *In the context of any regularized energy with a specified λ in Eq. (11), there exists a value $0 < \tau < 1$ in Eq.(10), ensuring that the diffusion process with the diffusivity in our method can yield a descent step on the energy across successive steps, symbolized as $E(Z^{(k+1)}, k; \delta) \leq E(Z^{(k)}, k-1; \delta)$, valid for each $k \leq 1$.*

The proof can refer to Appendix A. Theorem 1 elucidates the presence of an intrinsic optimal diffusivity function, fundamentally dependent on the distance between current states, such as $\|z_i^{(k)} - z_j^{(k)}\|_2$. This significant insight allows for a meticulous deconstruction of the traditionally implicit diffusion process, thereby enabling precise calculation of the diffusion matrix $S^{(k)}$ through a systematic feed-forward approach, starting from the initial states. In this way, it can enhance computational precision and clarity, ensuring a more robust and reliable implementation of the associated algorithms and methodologies.

Leveraging the crucial insights from Theorem 1, we are equipped to identify the appropriate diffusivity function, based on the differences in distances between corresponding states at each successive stage. This approach enables us to precisely model the diffusion process with clear accuracy, avoiding the need for repetitive computations. This method also facilitates the efficient determination of the diffusion matrix $S^{(k)}$, employing a streamlined feed-forward methodology that originates from the initial states. In this way, it demonstrates the practicality and benefits of Eq.(5).

6 EXPERIMENTS

In this section, we conduct comprehensive empirical experiments to assess the efficacy of the DiffusionE framework under both transductive and inductive learning settings.

6.1 Transductive setting

Datasets. We adopt six well-established datasets as shown in Table 1, ubiquitously utilized in KG completion tasks. Specifically, these datasets encompass Family [18], UMLS [18], WN18RR [6], FB15k237 [29], NELL995 [33], and YAGO3-10 [26].

Table 1: The statistics of the KG datasets used in our paper. Notice that the fact triplets in $\mathcal{F}$ are used to build the graph, and Q_{tra}, Q_{val}, Q_{tst} are the query triplets used for reasoning.

dataset	entity	relation	$\lvert\mathcal{F}\rvert$	$\lvert Q_{tra}\rvert$	$\lvert Q_{val}\rvert$	$\lvert Q_{tst}\rvert$
Family	3.0k	12	23.4k	5.9k	2.0k	2,8k
UMLS	135	46	5.3k	1.3k	569	633
WN18RR	40.9k	11	65.1k	21.7k	3.0k	3.1k
FB15k237	14.5k	237	204.1k	68.0k	17.5k	20.4k
NELL-995	74.5k	200	112.2k	37.4k	543	2.8k
YAGO3-10	123.1k	37	809.2k	269.7k	5.0k	5.0k

Baselines. For the experimental evaluation, we compare DiffusionE with a diverse array of baseline models including: (i) Non-GNN Methods such as ConvE [6], QuatE [40], RotatE [27], MINERVA [5], DRUM [24], RNNLogic [22], and RLogic [3]. (ii) GNN-Based Full Propagation Methods such as CompGCN [30], and Progressive Propagation Methods such as NBFNet [43] and RED-GNN [41]. Noteworthy is the omission of R-GCN [25], attributed to its inferior performance relative to CompGCN [30]. Moreover, as delineated by [41, 43], models such as GraIL [28], and CoMPILE [20].

Metrics. In line with previous research, we utilize two commonly used evaluation metrics: mean reciprocal ranking (MRR) and Hit@k, where K can be 1 and 10. Higher scores on these metrics indicate superior performance of the model.

Implementation. In our implementation, we explore a range of values for the propagation steps, denoted as L, spanning from 5 to 8. Additionally, we experiment with different numbers of sampled entities, denoted as K, which include values from the candidate set $\{100, 200, 500, 1000, 2000\}$. Alongside this, we vary the temperature values, denoted as τ, selecting from the set $\{0.5, 1.0, 2.0\}$. We ensure consistency in the range configurations of other hyperparameters with those used in the RED-GNN implementation [41].

Results. **Results**. Tables 1 and 2 provide a comprehensive overview of the performance of DiffusionE compared to established transductive reasoning methods. Notably, Graph Neural Network (GNN)-based methods show remarkable efficacy, outperforming non-GNN counterparts. This superiority can be attributed to their ability to capture both the structural and semantic complexities inherent in knowledge graphs, resulting in a more comprehensive and nuanced representation Within the realm of GNN-based methodologies, a nuanced differentiation is observed. Progressive propagation techniques, such as NBFNet and REDGNN, demonstrate enhanced performance compared to comprehensive propagation strategies like CompGCN. This disparity is particularly pronounced in extensive knowledge graphs such as WN18RR, FB15k237, NELL-995, and YAGO3-10, where progressive strategies exhibit notable advantages. DiffusionE stands out by achieving superior performance across various baseline methods, particularly in knowledge graphs such as WN18RR, NELL-995, and YAGO3-10. It even exhibits a slight yet noteworthy edge over NBFNet in the NELL-995 dataset. Furthermore, DiffusionE showcases adaptability and competitive performance in smaller knowledge graphs like Family and UMLS, highlighting its versatility and effectiveness. In summary, these results demonstrate the effectiveness of DiffusionE in transductive reasoning tasks across diverse knowledge graphs.

Table 2: The results of transductive setting on Family, UMLS, and WN18RR datasets. The best results are shown in bold while the second-best results are shown in the underline.

type	models	Family			UMLS			WN18RR		
		MRR	H@1	H@10	MRR	H@1	H@10	MRR	H@1	H@10
	ConvE	0.912	0.837	0.982	0.937	0.922	0.967	0.427	0.392	0.498
	QuatE	0.941	0.896	0.991	0.944	0.905	0.993	0.480	0.440	0.551
	RotatE	0.921	0.866	0.988	0.925	0.863	0.993	0.477	0.428	0.571
	MINERVA	0.885	0.825	0.961	0.825	0.728	0.968	0.448	0.413	0.513
non-GNN	DRUM	0.934	0.881	0.996	0.813	0.674	0.976	0.486	0.425	0.586
	RNNLogic	0.881	0.857	0.907	0.842	0.772	0.965	0.483	0.446	0.558
	RLogic	-	-	-	-	-	-	0.470	0.443	0.537
	NodePiece	0.892	0.837	0.971	0.844	0.736	0.968	0.459	0.432	0.543
	MorsE (RotatE)	0.938	0.886	0.994	0.815	0.679	0.978	0.493	0.435	0.596
	CompGCN	0.933	0.883	0.991	0.927	0.867	0.994	0.479	0.443	0.546
	NBFNet	0.989	0.988	0.989	0.948	0.920	**0.995**	0.551	0.497	**0.666**
GNN	ConGLR	0.982	0.971	0.969	0.952	0.923	0.981	0.532	0.473	0.654
	RED-GNN	**0.992**	0.988	**0.997**	0.964	0.946	0.990	0.533	0.485	0.624
	DiffusionE	0.990	**0.989**	0.992	**0.970**	**0.957**	0.992	**0.557**	**0.504**	0.658

Table 3: The results of transductive setting on FB15K237, NELL-995, and YAGO3-10 datasets. The best results are shown in bold while the second-best results are shown in the underline.

type	models	FB15K237			NELL-995			YAGO3-10		
		MRR	H@1	H@10	MRR	H@1	H@10	MRR	H@1	H@10
	ConvE	0.325	0.237	0.501	0.511	0.446	0.619	0.520	0.450	0.660
	QuatE	0.350	0.256	0.538	0.533	0.466	0.643	0.379	0.301	0.534
	RotatE	0.337	0.241	0.533	0.508	0.448	0.608	0.495	0.402	0.670
non-GNN	MINERVA	0.293	0.217	0.456	0.513	0.413	0.637	-	-	-
	DRUM	0.343	0.255	0.516	0.532	0.460	0.662	0.531	0.453	0.676
	RNNLogic	0.344	0.252	0.530	0.416	0.363	0.478	0.554	0.509	0.622
	RLogic	0.310	0.203	0.501	-	-	-	0.36	0.252	0.504
	CompGCN	0.355	0.264	0.535	0.463	0.383	0.596	0.421	0.392	0.577
	NBFNet	**0.415**	**0.321**	**0.599**	0.525	0.451	0.639	0.550	0.479	0.686
GNN	RED-GNN	0.374	0.283	0.558	0.543	0.476	0.651	0.559	0.483	0.689
	DiffusionE	0.376	0.294	0.539	**0.552**	**0.490**	**0.654**	**0.566**	**0.494**	**0.692**

6.2 Inductive setting.

Datasets. Following [28, 41], we utilize a total of twelve subsets, encompassing four distinct versions each, which are derived from established datasets: WN18RR, FB15k237, and NELL-995. Each subset is uniquely characterized by a disparate split between the training and test sets, ensuring a robust and comprehensive evaluation schema. For a detailed discourse on the specific splits and a nuanced presentation of the statistical attributes of each subset, one can refer to the comprehensive descriptions available in [28, 41].

Baselines. For our inductive setting evaluation, the comparative baselines are as follows. Specifically, reasoning methods, such as ConvE [6], QuatE [40], RotatE [27], MINERVA [5], and CompGCN [30], which predominantly rely on the learning of entity embeddings during training, are inherently unsuitable for this context and hence excluded. Our comparison pool comprises non-GNN methods like RuleN [21], NeuralLP [35], and DRUM [24], which emphasize rule learning devoid of entity embeddings. In the realm of GNN-based methods, our selection includes noteworthy models like GraIL [28], CoMPILE [20], REDGNN [41], and NBFNet [43]. Models such as RNNLogic [22] and RLogic [3], despite their relevance, are omitted from comparison due to the unavailability of essential resources like source codes and particular results in this setting. For a robust and comprehensive evaluation, we adopted the evaluation strategy from [41], prioritizing the ranking of target answer entities over a complete spectrum of negative entities, diverging from the approach in [28] that utilizes a subset of 50 randomly sampled negative entities.

Results. As shown in Table 4 and Table 5, constrained propagation techniques, represented by GraIL and CoMPILE, despite their inherent capabilities for inductive reasoning, do not perform as well as their progressive propagation counterparts, notably RED-GNN and NBFNet. The patterns internalized by GraIL and CoMPILE within their constrained operational scopes appear to falter in generalizing effectively to unfamiliar KGs. In contrast, DiffusionE achieves high performance across diverse datasets and various data split versions.

Table 4: Results for inductive setting (evaluated with Hit@10). The best results are shown in bold while the second-best results are shown in the underline.

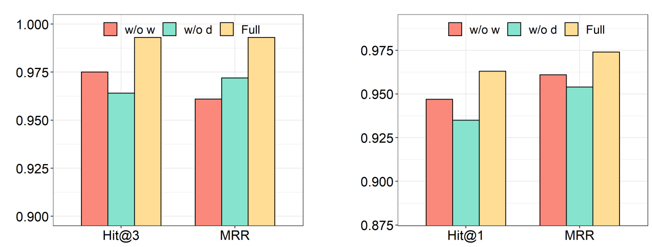
Methods	WN18RR				FB15K237				NELL-995			
	V1	V2	V3	V4	V1	V2	V3	V4	V1	V2	V3	V4
RuleN	0.730	0.694	0.407	0.681	0.446	0.599	0.600	0.605	0.760	0.514	0.531	0.484
Neural LP	0.772	0.749	0.476	0.706	0.468	0.586	0.571	0.593	0.871	0.564	0.576	0.539
DRUM	0.777	0.747	0.477	0.702	0.474	0.595	0.571	0.593	0.873	0.540	0.577	0.531
GraIL	0.760	0.776	0.409	0.687	0.429	0.424	0.424	0.389	0.565	0.496	0.518	0.506
CoMPILE	0.747	0.743	0.406	0.670	0.439	0.457	0.449	0.358	0.575	0.446	0.515	0.421
NBFNet	0.827	0.799	0.563	0.702	0.517	0.639	0.588	0.559	0.795	0.635	0.606	**0.591**
RED-GNN	0.799	0.780	0.524	0.721	0.483	0.629	0.603	0.621	0.866	0.601	0.595	0.556
DiffusionE	**0.857**	**0.843**	**0.605**	**0.737**	**0.525**	**0.661**	**0.627**	**0.625**	**0.873**	**0.660**	**0.620**	0.523

Table 5: Results for inductive setting (evaluated with MRR). The best results are shown in bold while the second-best results are shown in the underline.

Methods	WN18RR				FB15K237				NELL-995			
	V1	V2	V3	V4	V1	V2	V3	V4	V1	V2	V3	V4
RuleN	0.668	0.645	0.368	0.624	0.363	0.433	0.439	0.429	0.615	0.385	0.381	0.333
Neural LP	0.649	0.635	0.361	0.628	0.325	0.389	0.400	0.396	0.610	0.361	0.367	0.261
DRUM	0.666	0.646	0.380	0.627	0.333	0.395	0.402	0.410	0.628	0.365	0.375	0.273
GraIL	0.627	0.625	0.323	0.553	0.279	0.276	0.251	0.227	0.481	0.297	0.322	0.262
CoMPILE	0.577	0.578	0.308	0.548	0.287	0.276	0.262	0.213	0.330	0.248	0.319	0.229
RED-GNN	0.701	0.690	0.427	0.651	**0.369**	0.469	0.445	0.442	0.637	0.419	0.436	**0.363**
NBFNet	0.684	0.652	0.425	0.604	0.307	0.369	0.331	0.305	0.584	0.410	0.425	0.287
DiffusionE	**0.727**	**0.724**	**0.460**	**0.660**	0.310	**0.474**	**0.456**	**0.446**	**0.673**	**0.424**	**0.458**	0.308

This consistency underscores the effectiveness of DiffusionE in extrapolating the adaptively learned propagation paths to novel KGs, maintaining its effectiveness even in the presence of previously unseen entities during the training phase. These results demonstrate DiffusionE's robust inductive reasoning capabilities, standing as a testament to its generalized applicability and performance.

Ablation Study. In order to evaluate the effectiveness of our proposed method, we conducted an ablation study on different datasets. The results of this study are presented in Figure 2, which shows the performance comparison between different variants of our method, namely "w/o w" and "w/o d". Specifically, "w/o w" refers to the variant where the wave-based propagation is removed, while "w/o d" refers to the variant where the semantic-dependent diffusion is removed. From the results, it is evident that the experimental performance of all the ablation versions decreases compared to the original method. This shows the effectiveness of both the wave-based propagation and the semantic-dependent diffusion in DiffusionE.



(a) Family **(b) UMLS**

Figure 3: Performance Comparison between different variants of DiffusionE.

7 CONCLUSION

In contrast to existing GNN-based approaches that rely on predefined or sampled propagation paths, we introduce a new method termed DiffusionE by learning propagation paths through a diffusion mechanism during message propagation. This approach involves two key components: a wave-based propagation module that efficiently preserves nearby targets and maintains layer-wise connections with linear complexity among entities, and a semantics-based flowing module that identifies semantically related entities during the propagation process. Our empirical evaluation of various benchmark datasets considers both transductive and inductive KG reasoning scenarios. The results highlight the effectiveness of DiffusionE in achieving state-of-the-art performance on several datasets and demonstrate the superiority of DiffusionE from both practical and theoretical perspectives. Moreover, noticing that the large language model (LLM) embraces rich semantic information for entities, combining the DiffusionE and LLM may be a promising direction for enhancing the GCN framework in the context of KG reasoning.

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